

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

Electron Configuration

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: shells, subshells, Aufbau, orbital boxes and ion configurations.

A Retrieval

1. State the maximum number of electrons in the first, second and third shells for the first 20 elements as used in this course.

2. How many electrons can an s subshell and a p subshell hold?

3. What does the group number tell you about a Group 1 or Group 7 atom?

B Knowledge

4. Write shell configurations for N, Mg, Cl and K.

5. Write full subshell configurations for O, Al and Ca. Use superscripts, e.g. $1s^2 2s^2 2p^4$.

Orbital box diagrams — boxes and spin arrows drawn

Each box is one orbital. An up arrow and a down arrow are electrons of opposite spin.

Helium $1s^2$ 

Pauli: two electrons in one orbital must have opposite spin.

Nitrogen $2p^3$ 

Hund: one electron in each equal-energy orbital before pairing.

Oxygen $2p^4$ 

The fourth 2p electron pairs in one box. The other two stay unpaired.

6. On the orbital-box idea above, draw boxes and arrows for nitrogen $2p^3$ and oxygen $2p^4$. State Hund's rule in one sentence.

C Application

7. Write the configuration of O^{2-} , Al^{3+} and Cl^- in both shell and subshell form.

8. A student writes calcium as 2,8,10. Correct the configuration and explain the error.

9. Explain, using electrons, why magnesium forms a $2+$ ion.

D Exam-style practice

10. Write the electron configuration of Mg^{2+} and explain why the ion is stable. [3]

11. Explain why oxygen typically forms a $2-$ ion. [3]

12. Identify the block (s, p or d) for sodium, fluorine and iron, and justify one of them. [3]

E Challenge

13. Write [Ne] shorthand configurations for phosphorus and for P^{3-} . Explain the relationship between them.
